LECTURE 1

PRESENTATION

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LECTURE 1

GEOMETRIC AND PHYSICAL PROBLEMS LEADING TO DIFFERENTIAL EQUATIONS. THE CONCEPT OF A FIRST-ORDER ORDINARY DIFFERENTIAL EQUATION.

Purpose of the lecture: to consider geometric and physical problems leading to differential equations, examples, to introduce the concept of an ordinary first-order differential equation solvable with respect to the derivative, and its solutions.

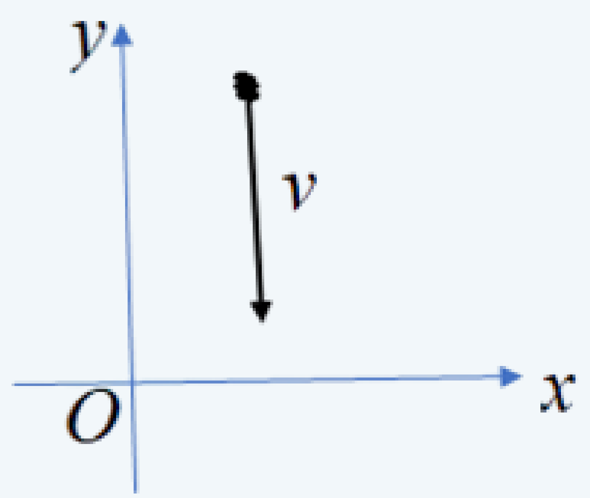
1.1 GEOMETRIC AND PHYSICAL PROBLEMS LEADING TO DIFFERENTIAL EQUATIONS, EXAMPLES

When solving any practical problem using algebra, the unknown quantity is denoted by x, and an algebraic equation is formed according to the conditions of the problem. By solving this equation, we find the unknown value of x. However, many practical problems in geometry, mechanics, physics, and other fields of knowledge require finding a function y(x) that satisfies the requirements arising from the conditions of the problem. Let us give some examples.

EXAMPLE 1.

As is known from physics, the velocity of a freely falling body at time t is equal to $v = gt$, where g = 9.81 m/s2 is the acceleration due to gravity. Find the distance of the body from the Earth at time t if it began to fall from a height of y0.

Solution. Let's choose the origin at a point on the Earth's surface and direct the Oy axis vertically upward.



Then the velocity vector v will be directed vertically downward, and its projection onto the Ou axis will be equal to $(-gt)$. On the other hand, velocity is equal to the derivative of displacement with respect to time dy/dt. Equating both expressions for velocity, we obtain the equation:

$dy\ /\ dt = -gt$ **or** $dy = -gt*dt$

Integrating both sides of this equation, we obtain:

$$y = -g \int t dt$$

or

$$y = -\frac{gt^2}{2} + C.$$

We have obtained a set of functions from which we need to find one that

satisfies the initial condition: when t = 0, y = y0. Substituting these values into equation (1.1), we find C = y0. Thus, at time t, a freely falling body is at a distance from the Earth:

y = -gt/2+y0

Note that in this example, the solution is not a number, but a **function**

EXAMPLE 2

WRITE AN EQUATION FOR A CURVE PASSING THROUGH THE POINT WITH COORDINATES $M(-2; 3)$ IF THE SLOPE OF THE TANGENT TO THIS CURVE AT ANY POINT ON IT IS EQUAL TO THE X-COORDINATE OF THAT POINT.

Solution. According to the geometric meaning of the derivative, the angular coefficient of the tangent to a given curve at a given point $M(x, y)$ is equal to $\frac{dy}{dx}$

On the other hand, according to the condition of the problem, this angular coefficient
is equal to the abscissa of the point of contact. Thus, we have:

$$\frac{dy}{dx} = x,$$

from here

$$dy = xdx.$$

Integrating both parts of the equation, we find:

$$y = \frac{x^2}{2} + C.$$

This equation is a family of parabolas, since C is an arbitrary constant that can take any numerical value. From this family of parabolas, we need to select the curve that passes through the given point $M(-2; 3)$. Substituting the coordinates of the point $M(-2; 3)$ into the last equation, we obtain:

$$3 = (-2)^2/2 + C$$

from which C=1. With this value of C, we obtain the function:

$$y = \frac{x^2}{2} + 1,$$

which satisfies all the conditions of the given problem, i.e., it is the desired function.

We have considered problems of a geometric and physical nature that lead to differential equations.

QUESTIONS FOR SELF-ASSESSMENT

- **1**. HOW IS THE GENERAL FORM OF A FIRST-ORDER EQUATION DETERMINED**?**
- **2**. WHAT IS THE NAME OF THE SOLUTION TO A DIFFERENTIAL EQUATION THAT HAS A SET OF FUNCTIONS, FROM WHICH ONE MUST FIND THE ONE THAT SATISFIES THE INITIAL CONDITION**?**
- **3**. IF A FAMILY OF PARABOLAS IS GIVEN AND WE NEED TO SELECT A CURVE PASSING THROUGH A GIVEN POINT M(-2; 3), THEN THE COORDINATES OF THE GIVEN POINT ARE **…** .
- **4**. WHAT IS THE NAME OF THE EQUATION LINKING THE INDEPENDENT VARIABLE X, THE UNKNOWN FUNCTION Y, AND ITS DERIVATIVES OR DIFFERENTIALS OF VARIOUS ORDERS**?**
- **5**. WHAT IS THE NAME OF THE ORDER OF THE HIGHEST DERIVATIVE INCLUDED IN A DIFFERENTIAL EQUATION**?**
- **6**. HOW IS A FIRST-ORDER EQUATION SOLVED WITH RESPECT TO THE DERIVATIVE DEFINED**?**

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